Nestack Cloud & System Architecture Overview
Page 1: Core Infrastructure and Ingestion Services

1.1 Introduction
Nestack builds resilient enterprise data pipelines designed to process high-throughput unstructured docum

1.2 Text Ingestion Pipeline
The document ingestion subsystem relies on PyMuPDF (fitz) to accurately extract text streams while main

1.3 Text Chunking Strategy
Raw text is split into fixed-size overlapping chunks. A default window size of 500 characters with a 100-cha

2.1 Sentence Transformers & Dense Embeddings

For dense semantic vector representation, Nestack utilizes the all-MiniLM-L6-v2 sentence transformers mo

2.2 ChromaDB Persistence & Indexing

ChromaDB operates as the primary vector store. Ingested vectors are indexed using Hierarchical Navigab

2.3 High-Throughput Batch Upsert

Embeddings are computed in batches of 32 to 64 items to optimize tensor memory usage. ChromaDB ups

3.1 Semantic Search vs Chatbots

Nestack retrieval endpoints implement pure semantic search. Under no circumstances should the system i

3.2 FastAPI Query Endpoint Specification

The service exposes a POST /query endpoint receiving a JSON payload containing the query string. The s

3.3 Response JSON Schema

The response must conform to: chunk_text (string), page_number (integer), score (float). A higher score in